A
TA-III Report
On
"Interview Preparation Chatbot for Technical Roles"

By

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[2025-26]

1. Introduction

In today's education system, students often need a large number of practice questions to prepare effectively for exams, tests, and competitive assessments. However, creating quality questions manually from study material takes a lot of time and effort. Many students also struggle to find a simple tool that can automatically generate useful questions from their notes or textbooks.With the advancement of **Artificial Intelligence (AI)** and **Natural Language Processing (NLP)**, it is now possible to build intelligent systems that can analyze text and automatically generate meaningful questions and answers. AI-powered educational tools can help students by converting study material into practice questions in different formats, making learning more interactive and efficient.

The **AI-Based Question Generator for Exam Practice** is designed to help students generate exam-oriented questions automatically from study material. The system allows users to either upload a **PDF document** or manually enter notes or a topic. Based on the selected **question type**, **difficulty level**, and **number of questions**, the system generates questions using a **local Large Language Model (LLM)**.

The main objective of this system is to provide an intelligent and interactive platform where students can quickly create practice questions, test their knowledge, and improve their exam preparation without depending on external APIs or internet-based AI services.

2. System Overview

The **AI-Based Question Generator** is an AI-powered application that helps students create practice questions from their study material in an easy and interactive way. The system is developed as a **Streamlit-based web application** where users can provide study content and instantly generate questions.

The system supports **two input methods**:

- **PDF Upload** – The user can upload a study material PDF, and the system extracts text from the document.
- **Manual Text Input** – The user can type or paste notes, a topic, or study material directly into the application.

  After providing the input, the user can configure **:**

- **Question Type** (MCQ, True/False, Short Answer, Long Answer, Mixed)
- **Difficulty Level** (Easy, Medium, Hard)
- **Number of Questions**

Once the user clicks the **Generate Questions** button, the system sends the study material and settings to the AI model. The AI model generates only the questions first, allowing the student to attempt them independently.

This approach makes the application more useful for **self-assessment**, **revision**, and **exam preparation**.

3. Technology Stack

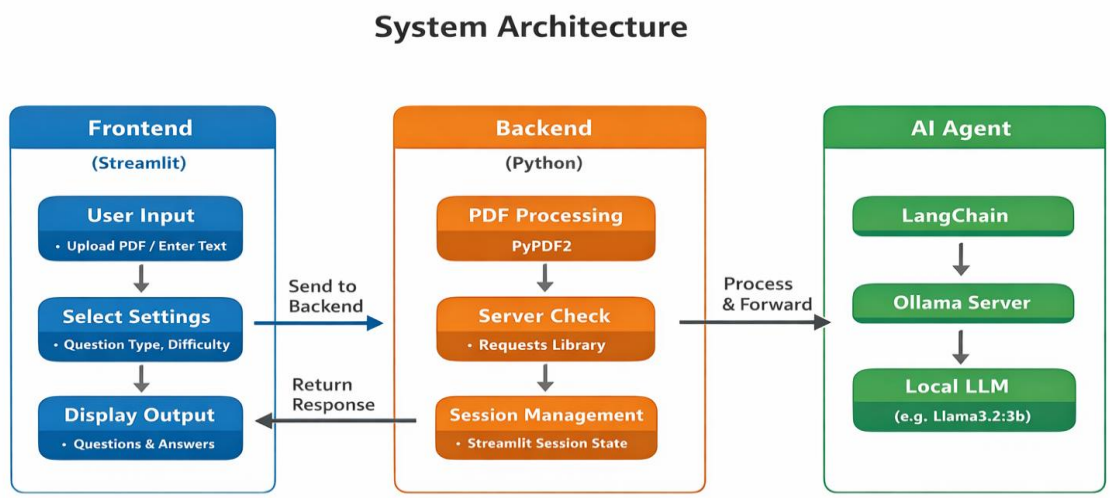
The frontend of the project is built using Streamlit**.** It provides an interactive web interface where users can upload PDFs, enter text, select question settings, and view/download generated questions and answers.

The backend is developed using Python**.** It handles user input, extracts text from PDFs using PyPDF2, checks the Ollama server using the Requests library, and manages session data using Streamlit Session State**.**

The AI agent is built using LangChain, Ollama, and a local LLM like llama3.2:3b. It processes the study material and generates questions and answers based on the user's selected settings

4. System Architecture

The system follows a **modular AI-based architecture** where the user interacts with the Streamlit interface, and the backend processing is handled in Python with the help of AI and PDF extraction modules.



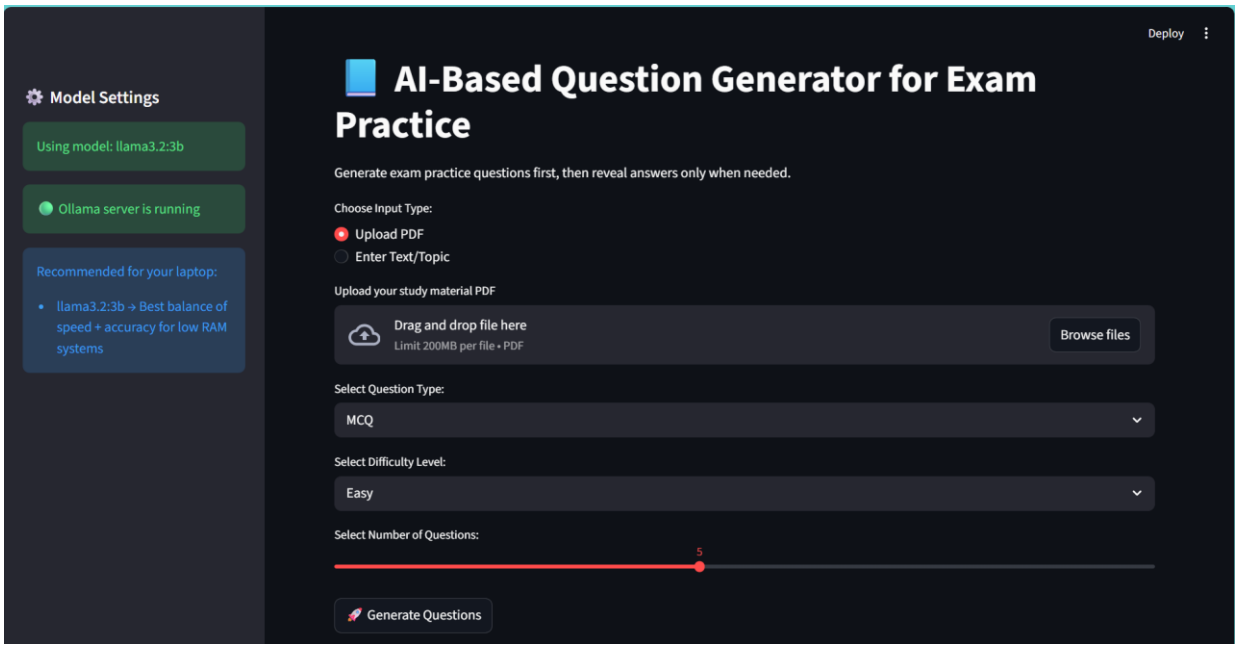
- Backend processes request and sends it to AI module
- AI module generates answer and evaluation score
- Response is displayed in the chat interface

5. Implementation

The AI-Based Question Generator is implemented as a Streamlit web application using Python. The system allows users to either upload a PDF file or manually enter study material as text. If a PDF is uploaded, the PyPDF2 library extracts the text content from the document. The user then selects the question type, difficulty level, and number of questions through the interface.

After receiving the input, the backend checks whether the Ollama server is running using the Requests library. Once the server is available, the study material and selected settings are sent to the LangChain + Ollama module, which interacts with the local LLM (such as llama3.2:3b) to generate questions. The system first displays only the generated questions, and when the user clicks Show Answers, it generates answers separately. Streamlit Session State is used to store questions and answers during the session, and the output can also be downloaded as a text file.

6. Results and Output



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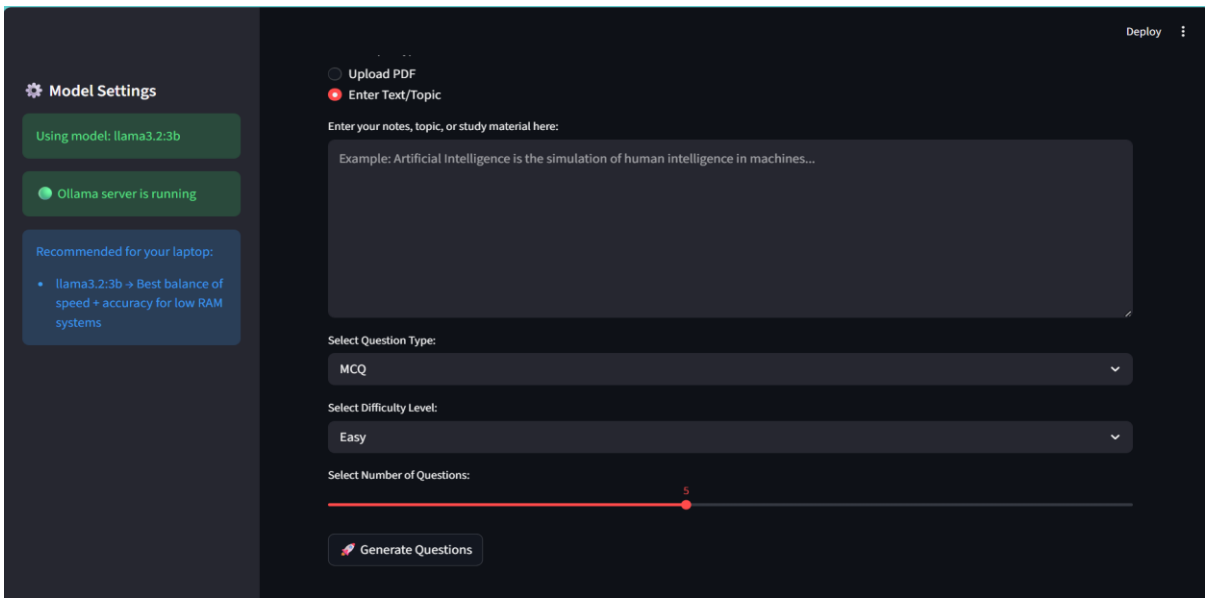

The results show that the system successfully simulates and helps users improve their technical understanding.

7. Conclusion

The AI-Based Question Generator for Exam Practice is an effective AI-powered educational tool that helps students generate questions automatically from study material. It reduces the effort required to manually create practice questions and provides an interactive way to prepare for exams.

The system supports both PDF-based input and manual text input, making it flexible for different study methods. By using a local LLM through Ollama, the application works without relying on external APIs, which improves privacy and reduces dependency on internet-based services.

8. References

- Python Official Documentation
- Streamlit Official Documentation
- PyPDF2 Official Documentation
- LangChain Official Documentation